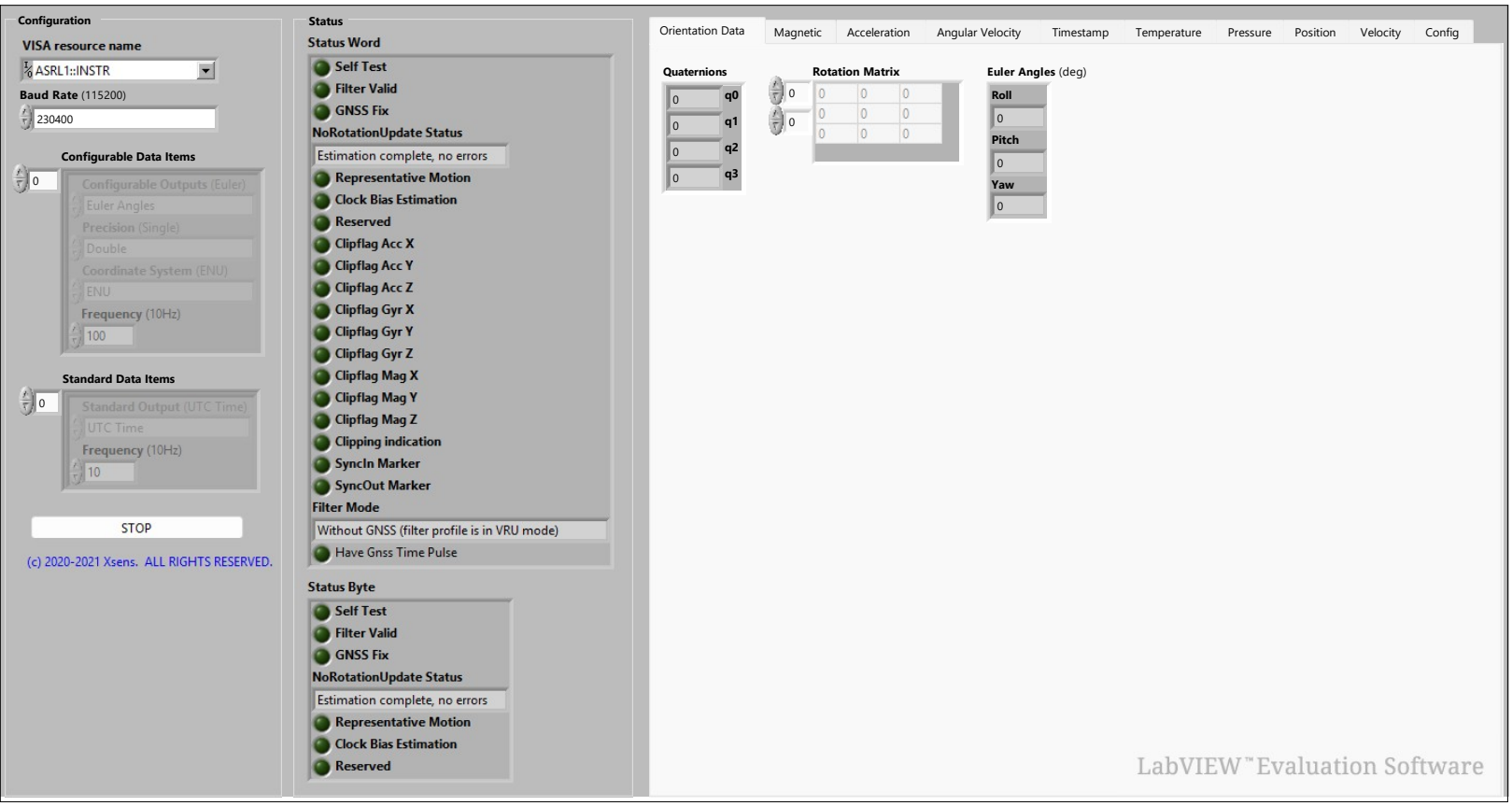
xsens MTi Series Stream All Data.vi
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xsens MTi Series Stream All Data.vi

Configure the MT to stream all possible data and display it to the user.



Stop
Stop the VI from running.





VISA resource name
Specifies which instrument to use. Instruments that are powered on and connected to your computer are visible in the <B>VISA resource name</B> pull-down menu. Select the resource name from the <B>VISA resource name</B> pull-down menu. If your instrument does not appear in this list, enter the instrument descriptor using the following format:
Interface Type[board index]::Address::INSTR.

<B>Examples:</B>
<B>GPIB:</B> GPIB::22::INSTR, GPIB board 0, primary address 22
<B>Serial:</B> COM1
<B>TCP/IP Instr:</B> TCPIP::10.0.32.48::INSTR
<B>TCP/IP Socket:</B> TCPIP::10.0.32.45::3800::SOCKET
<B>USB Instr:</B> Selectable from VISA Resource Name pull-down menu
<B>USB Raw:</B> You must first link VISA to non-USBTMC instruments by using the VISA Driver Development Wizard
<B>VISA Alias:</B> "SampleInstr" - Alias names are configured in MAX

Refer to the LabVIEW Help for more information about specifying a VISA resource name.



Tab Control



Configurable Data Items
An array of the configurable data items to be streamed by the MTi.



Data Item
A data packet that can be returned from the MT



Configurable Outputs (Euler)
The output that will be streamed from the MTi when in measurement mode.

Valid Values:
Temperature
Quaternion
Rotation Matrix
Euler Angles
Delta V
Acceleration
Free Acceleration
Rate Of Turn
Magnetic Field
Delta Q
Baro Pressure
Altitude Ellipsoid
Position ECEF
Lat & Lon
Velocity

Default Value:
Euler Anlgles





Precision (Single)
The output format of the MTi.

Valid Values:
Single
12.20
16.32
Double

Default Value:
Single



Coordinate System (ENU)
ENU - East-North-Up coordinate system
NED - North-East-Down coordinate system
NWU - North-West-Up

Valid Values:
ENU
NED
NWU

Default Value:
ENU



Frequency (10Hz)
The rate at which data is sent from the MTi.

Valid Range:
Min: 1Hz
Max: 65535Hz

Default Value:
10



Standard Data Items
An array of the standard data items to be streamed by the MTi.



Standard Data Item
A data item that can't be configured.





Standard Output (UTC Time)
The output that will be streamed from the MTi when in measurement mode.

Valid Values:
UTC Time
Packet Counter
Sample Time Fine
Sample Time Coarse
Status Byte
Status Word

Default Value:
UTC Time



Frequency (10Hz)
The rate at which data is sent from the MTi.

Valid Range:
Min: 1Hz
Max: 65535Hz

Default Value:
10



Baud Rate (115200)
Specifies the rate at which data is transmitted across the serial interface.

Valid Values:
921600
460800
230400
115200
57600
38400
28800
19200
14400
9600
4800

Default Value:
115200



Rotation Matrix
A Rotation Matrix that represents the orientation of the MT.



Real Matrix Element

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Euler Angles (deg)
The three Euler angles that represent the orientation of the MT.

Roll
Returns the Roll of the device with a range of -180.0 to +180.0 degrees.

Pitch
Returns the Pitch of the device with a range of -90.0 to +90.0 degrees.

Yaw
Returns the Yaw of the device with a range of -180.0 to +180.0 degrees.

Quaternions
Quaternions cluster

q0
Returns the Roll of the device with a range of -180.0 to +180.0 degrees.

q1
Returns the Pitch of the device with a range of -90.0 to +90.0 degrees.

q2
Returns the Yaw of the device with a range of -180.0 to +180.0 degrees.

q3
Returns the Yaw of the device with a range of -180.0 to +180.0 degrees.

Magnetic Field (a.u.)
Contains the calibrated magnetic field value in X, Y, and Z axes

magX
The calibrated magnetic field value in X

magY
The calibrated magnetic field value in Y

magZ
The calibrated magnetic field value in Z

Delta V (m/s)
Contains the delta velocity value of the SDI output

v.x
Delta velocity in X

v.y
Delta velocity in Y

v.z
Delta velocity in Z

Acceleration (m/s^2)
The calibrated acceleration vector in X, Y, and Z axes

accX
Contains the calibrated acceleration vector in X axes

accY
Contains the calibrated acceleration vector in Y axes

accZ
Contains the calibrated acceleration vector in Z axes

Free Acceleration (m/s^2)
Contains the free acceleration vector in X, Y, and Z axes

freeaccX
Contains the free acceleration vector in X

freeaccY
Contains the free acceleration vector in Y

freeaccZ
Contains the free acceleration vector in Z

Delta Q
Contains the delta quaternion value of the SDI output

q0
Delta quaternion 0

q1
Delta quaternion 1

q2
Delta quaternion 2

q3
Delta quaternion 3

Rate Of Turn (rad/s)
Contains the calibrated rate of turn vector in X, Y, and Z axes

gyrX
Contains the calibrated rate of turn vector in X

gyrY
Contains the calibrated rate of turn vector in Y

gyrZ
Contains the calibrated rate of turn vector in Z

UTC Time
The UTC Time from the GPS sensor.

Sample Time Fine
Contains the timestamp of an output (sample) expressed in 10kHz ticks.

Sample Time Coarse
Contains the timestamp of an output (sample) expressed in seconds.

Packet Counter
This packet contains the packet counter. This counter is incremented with every generated MTData2 message

Temperature (Degrees Celsius)
The internal temperature of the sensor in degrees Celsius.

Baro Pressure (Pa)
The internal temperature of the sensor in degrees Celsius.

Altitude Ellipsoid (m)

LatLon

- **Latitude**
- **Longitude**

Velocity (m/s)
Contains the calibrated rate of turn vector in X, Y, and Z axes

- **velX**
  Contains the calibrated rate of turn vector in X
- **velY**
  Contains the calibrated rate of turn vector in Y
- **velZ**
  Contains the calibrated rate of turn vector in Z

Status Byte
Status contains flags that represent the status and estimated validity of the output of the MT.

- **Self Test**
  This flag indicates if the MT passed the latest self test (RunSelftest).
- **Filter Valid**
  This flag indicates if input into the XKF orientation filter is reliable and / or complete. If for example the measurement range of internal sensors is exceeded, orientation output cannot be reliably estimated and the XKF flag will drop to 0.
- **GNSS Fix**
  This flag indicates if the GPS unit has a proper fix.
- **NoRotationUpdate Status**
  This flag indicates the status of the no rotation update procedure in the filter after the SetNoRotation message has been sent. See User Manual for more details.

  1. No Error - Estimation complete, no errors
  2. Samples Rejected - Estimation complete, some samples rejected (sticky)
  3. Rotation Detected - Error: Rotation detected, procedure not started (sticky)
  4. Running with no rotation - Running with no rotation assumption

- **Representative Motion**
  The GPS time is synced with the internal clock. For this you need to wait for a max of 12.5 minutes to get the leap seconds.
- **Clock Bias Estimation**
  The internal clock of the MTi-G 700 is synced with the GPS 1 pps pulse
- **Reserved**
  This Flag is not used

Status Word
Status contains flags that represent the status and estimated validity of the output of the MT.

Self Test
This flag indicates if the MT passed the latest self test (RunSelftest).

Filter Valid
This flag indicates if input into the XKF orientation filter is reliable and / or complete. If for example the measurement range of internal sensors is exceeded, orientation output cannot be reliably estimated and the XKF flag will drop to 0.

GNSS Fix
This flag indicates if the GPS unit has a proper fix.

NoRotationUpdate Status
This flag indicates the status of the no rotation update procedure in the filter after the SetNoRotation message has been sent. See User Manual for more details.

1. No Error - Estimation complete, no errors
2. Samples Rejected - Estimation complete, some samples rejected (sticky)
3. Rotation Detected - Error: Rotation detected, procedure not started (sticky)
4. Running with no rotation - Running with no rotation assumption

Representative Motion
The GPS time is synced with the internal clock. For this you need to wait for a max of 12.5 minutes to get the leap seconds.

Clock Bias Estimation
The internal clock of the MTi-G 700 is synced with the GPS 1 pps pulse

Reserved
This Flag is not used

Clipflag Acc X
If true, an out of range acceleration on the X axis is detected

Clipflag Acc Y
If true, an out of range acceleration on the Y axis is detected

Clipflag Acc Z
If true, an out of range acceleration on the Z axis is detected

Clipflag Gyr X
If true, an out of range angular velocity on the X axis is detected.

Clipflag Gyr Y
If true, an out of range angular velocity on the Y axis is detected

Clipflag Gyr Z
If true, an out of range angular velocity on the Z axis is detected.

Clipflag Mag X
If true, an out of range magnetic field on the X axis is detected

Clipflag Mag Y
If true, an out of range magnetic field on the Y axis is detected

Clipflag Mag Z
If true, an out of range magnetic field on the Z axis is detected

Clipping indication
This flag indicates going out of range of one of the sensors (is set when one or more bits from 8: 16 are set)

SyncIn Marker
When a SyncIn is detected, this is true

SyncOut Marker
When SyncOut is active this is true

Filter Mode
This flag indicates the status of the no rotation update procedure in the filter after the SetNoRotation message has been sent. See User Manual for more details.

1. No Error - Estimation complete, no errors
2. Samples Rejected - Estimation complete, some samples rejected (sticky)
3. Rotation Detected - Error: Rotation detected, procedure not started (sticky)
4. Running with no rotation - Running with no rotation assumption

Have Gnss Time Pulse
When SyncOut is active this is true

Firmware Revision
The firmware revision of the MT. (Major.Minor.Rev)

Product Code
The product code of the MT set by the manufacturer.

Device ID
The serial number of the MT.

Self Test
The result of the self-test for the accelerometer, gyro and magnetometer. Pass = True. Fail = False.

Acc X
X Accelerometer sensor self-test result. TRUE = Passed

Acc Y
Y Accelerometer sensor self-test result. TRUE = Passed

Acc Z
Z Accelerometer sensor self-test result. TRUE = Passed

Gyro X
X Gyroscope sensor self-test result. TRUE = Passed

Gyro Y
Y Gyroscope sensor self-test result. TRUE = Passed

Gyro Z
Z Gyroscope sensor self-test result. TRUE = Passed

Mag X
X Magnetometer sensor self-test result. TRUE = Passed

Mag Y
Y Magnetometer sensor self-test result. TRUE = Passed

Mag Z
Z Magnetometer sensor self-test result. TRUE = Passed

UTC Time Status
UTC Time Status shows the validity of the UTC Time.

Valid Time of Week
Time of Week is Valid if true.

Valid Week Number
Week Number is Valid if true.

Valid UTC
UTC is Valid if true.

UTC Time
Coordinated Universal Time (UTC) from the MT.

